

Chapter 1: Introduction

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Game theory is a misnomer for **MULTIPERSON DECISION THEORY**: it develops tools, methods, and language for analyzing the choices of agents whose payoffs depend on each other's actions. A player's preference over his own actions depends on his beliefs about what the others do; what the others do depends on their beliefs about what each player does; and so on, ad infinitum. This chapter illustrates the main reasoning moves of the course on a handful of small games. The names will return in later chapters with full definitions; here the point is to see them at work.

1.1 A game solved by iterated reasoning

Throughout, payoffs are written as ordered pairs (u_1, u_2) : the first entry is Player 1's payoff, the second is Player 2's. Player 1 chooses the row, Player 2 chooses the column, and they choose simultaneously. Consider the following game.

$1 \backslash 2$	L	m	R
T	1, 1	0, 2	2, 1
M	2, 2	1, 1	0, 0
B	1, 0	0, 0	-1, 1

Assume each player knows the strategies and payoffs, each knows that the other knows, and so on without limit — call this **COMMON KNOWLEDGE** of the game.

Player 1 compares M and B . For every column, M pays more: $2 > 1$ under L , $1 > 0$ under m , $0 > -1$ under R . So B is worse than M no matter what Player 2 does, and Player 1 should not play B . Now look at the game from Player 2's side. None of L, m, R is unambiguously best: R beats m if and only if Player 1 plays B , and otherwise m beats R .

But Player 2, knowing Player 1 reasons as above, can rule out B . With B gone, m beats R everywhere, so R is ruled out too. Returning the favor, Player 1 knows Player 2 rules out R , and in the remaining 2×2 game M beats T . So Player 1 plays M , and Player 2, expecting M , plays L . The predicted outcome is (M, L) with payoff $(2, 2)$.

ITERATED ELIMINATION — STEP THROUGH

At each step, one strategy is strictly worse than another no matter what the opponent plays. Strike it and rerun.

Reset

Next step

STEP 0 — ORIGINAL GAME

No strategies eliminated yet. Press **Next step** to begin.

1\2	L	m	R
T	1, 1	0, 2	2, 1
M	2, 2	1, 1	0, 0
B	1, 0	0, 0	-1, 1

EXERCISE 1. *The reasoning above assumes a lot about what each player assumes about the other's reasoning. What exactly are those assumptions? How would the analysis change if players act rationally but each assumes the other plays a uniformly random strategy?*

1.2 When iterated reasoning gives no answer: coordination

The previous game admits a sharp prediction. Most games do not. Suppose you want to meet a friend in one of two equally good places, and you cannot communicate. This is the **PURE COORDINATION GAME**.

1\2	$Left$	$Right$
Top	1, 1	0, 0
$Bottom$	0, 0	1, 1

Player 1 prefers Top if he expects Left and Bottom if he expects Right; symmetrically for Player 2. Iterated elimination removes nothing. Asking "what does the other do?" recurses without progress.

One way out is to ask for **STABLE** profiles — profiles such that no player wants to deviate given the others' play.

DEFINITION. NASH EQUILIBRIUM. A strategy profile $(s_1^*, s_2^*, \dots)$ is a Nash equilibrium if, for each player i , s_i^* maximizes i 's payoff given that the other players play s_{-i}^* . Equivalently: no player has a profitable unilateral deviation.

In the pure coordination game both (Top, Left) and (Bottom, Right) are Nash equilibria. (Bottom, Left) and (Top, Right) are not: in each case, both players want to switch.

Battle of the Sexes

Usually preferences over outcomes disagree as well, and coordination must happen against a conflict. The classical **BATTLE OF THE SEXES** mixes the two.

$1 \backslash 2$	<i>Left</i>	<i>Right</i>
<i>Top</i>	2, 1	0, 0
<i>Bottom</i>	0, 0	1, 2

Both players want to coordinate, but Player 1 prefers (Top, Left) while Player 2 prefers (Bottom, Right). The Nash equilibria are again the two "matched" cells.

1.3 Sequential moves and backward induction

So far the players have moved simultaneously. If one player observes the other's action before choosing, the structure of the game — and its predictions — can change. Imagine the Battle of the Sexes played sequentially: Player 1 commits first, Player 2 sees the move, then Player 2 chooses. In the tree below the strategies are written B and O (think of "Bach" and "Opera"); B corresponds to Top/Left in the simultaneous matrix, O to Bottom/Right.

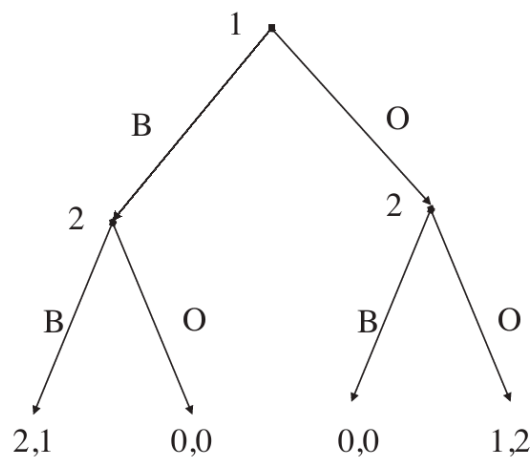


Figure 1.1: Battle of Sexes with sequential moves

Solve from the leaves up. If Player 1 played B , Player 2 picks B for a payoff of 1 instead of 0. If Player 1 played O , Player 2 picks O for 2 instead of 0. Anticipating this, Player 1 plays B — payoff 2 rather than 1 — and the outcome is (B, B) with payoffs $(2, 1)$. This solution method is called **BACKWARD INDUCTION**: work from the terminal nodes upward, replacing each subtree by its predicted payoff.

Now compare payoffs. In the simultaneous Battle of the Sexes there were two equilibria, giving Player 2 either 1 or 2. In the sequential version, Player 2 gets 1 for sure. Giving Player 2 better information about Player 1's move **HURT** Player 2. When it is common knowledge that a player has some piece of information, the player may prefer not to have it. The general phenomenon — information can hurt — recurs throughout the course.

EXERCISE 2. Suppose Player 2 sees Player 1's move only with probability $\pi < 1$, independent of what Player 1 chose. What is a reasonable equilibrium? By the end of the course you will be able to formalize this and compute it.

A second reading of the sequential game: Player 1 can **COMMIT** in a way Player 2 cannot. Commitment is power. Forcing yourself to act first, while the other side merely reacts, can be worth more than retaining flexibility.

EXERCISE 3. Extend the sequential game so that, after Player 2 chooses, Player 1 gets a chance to switch his action; then the game ends. What is the reasonable outcome? What if switching costs Player 1 c utiles?

1.4 Forward induction

Suppose that before playing the Battle of the Sexes, Player 1 may take an outside option that pays $(3/2, 3/2)$ to both players. If he plays, Player 2 sees that he chose to play and they enter the 2×2 Battle of the Sexes.

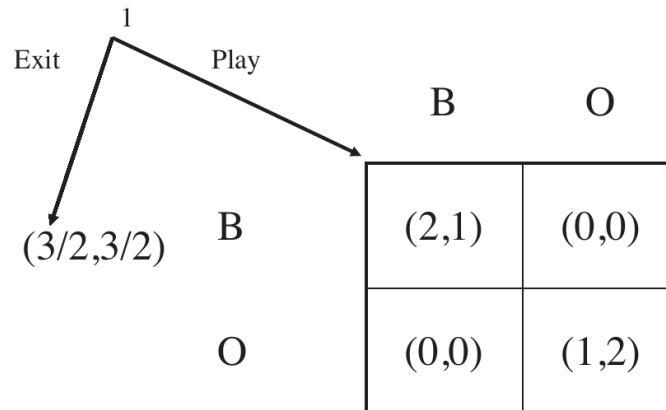


Figure 1.2: Battle of Sexes with exit option

Two stable patterns of play exist. In the first, Player 1 exits because he expects Player 2 to insist on (O, O) inside the subgame, leaving him with only 1 — so $3/2$ is better. In the second, Player 1 plays into the subgame and they coordinate on (B, B) , giving him 2.

The first outcome is questionable. When Player 2 is asked to move inside the subgame, she **KNOWS** Player 1 gave up the certain $3/2$. She must conclude he is not aiming at the (O, O) cell, which pays him only 1. So he must be aiming at (B, B) , and she should play B . Anticipating this, Player 1 plays into the subgame and they coordinate on (B, B) . This style of reasoning is called **FORWARD INDUCTION**: a deviation off the supposed equilibrium path is read as a signal about how the deviator intends to play later.

1.5 Three games that recur

Prisoners' Dilemma

1\2	<i>Cooperate</i>	<i>Defect</i>
<i>Cooperate</i>	5, 5	0, 6

<i>Defect</i>	6, 0	1, 1
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Two suspects, two rooms, no firm evidence. Each can keep quiet (cooperate) or confess (defect). For each player, Defect strictly beats Cooperate against either action of the opponent, so both defect and the outcome is $(1, 1)$ — even though both prefer $(5, 5)$. Individually rational play makes both worse off.

Hawk–Dove

$1 \backslash 2$	<i>Hawk</i>	<i>Dove</i>
<i>Hawk</i>	$\frac{V-C}{2}, \frac{V-C}{2}$	$V, 0$
<i>Dove</i>	$0, V$	$\frac{V}{2}, \frac{V}{2}$

Two animals contest a resource of value V . **HAWK** fights; **DOVE** yields. Two hawks meeting share a fight of cost C , so each gets $(V - C)/2$. A hawk meeting a dove takes the whole resource; two doves split it.

If $V > C$, fighting is cheap, Hawk strictly dominates Dove, and the game is essentially a Prisoners' Dilemma — both fight. If $V < C$, fighting is costly, and each player wants Hawk against a Dove and Dove against a Hawk; the game becomes a coordination problem with conflict, structurally a version of Chicken and the prototype for the class of **WARS OF ATTRITION**.

Investment game

$1 \backslash 2$	<i>Invest</i>	<i>Don't</i>
<i>Invest</i>	θ, θ	$\theta - c, 0$
<i>Don't</i>	$0, \theta - c$	$0, 0$

Two parties decide simultaneously whether to invest at cost c . Investment pays θ when matched and $\theta - c$ when not. Think of a worker considering education and a firm considering a technology that requires educated workers. The "reasonable" outcomes depend on the relative sizes of θ and c — and on what each side knows about the other's

parameters. Information about the opponent's payoff matters as much as the payoff itself, a theme the second half of the course makes precise.

PREVIOUS

NEXT

Chapter 2: Decision Theory

Based on Chapter 1 of [14.12 Economic Applications of Game Theory](#) (MIT OpenCourseWare, Fall 2025).